

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{36} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,0}[m_1, m_q^p] + \frac{1}{18} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,-1}[m_1, m_q^p] - \frac{1}{36} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} \right. \\
& LF_{4,1,-2}[m_1, m_q^p] - \frac{3}{4} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,0}[m_2, m_q^p] + \frac{3}{2} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,-1}[m_2, m_q^p] - \\
& \frac{3}{4} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{4,1,-2}[m_2, m_q^p] - \frac{4}{3} g_3^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,0}[m_3, m_q^p] + \\
& \frac{8}{3} g_3^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,-1}[m_3, m_q^p] - \frac{4}{3} g_3^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{4,1,-2}[m_3, m_q^p] - \\
& \frac{1}{2} \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{2,1,0}[\tilde{\mu}, m_d^r] + \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{3,1,-1}[\tilde{\mu}, m_d^r] - \\
& \frac{1}{2} \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{4,1,-2}[\tilde{\mu}, m_d^r] - \frac{1}{2} y_d^{ri2} \overline{y_u}^{pi1} \overline{y_u}^{rs} y_u^{ps} LF_{2,1,0}[\tilde{\mu}, m_u^s] + \\
& y_d^{ri2} \overline{y_u}^{pi1} \overline{y_u}^{rs} y_u^{ps} LF_{3,1,-1}[\tilde{\mu}, m_u^s] - \frac{1}{2} y_d^{ri2} \overline{y_u}^{pi1} \overline{y_u}^{rs} y_u^{ps} LF_{4,1,-2}[\tilde{\mu}, m_u^s] - \\
& \frac{1}{36} m_1 g_1^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{2,1,1,0}[m_1, m_d^{i2}, m_q^p] + \frac{1}{72} m_1 g_1^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{2,2,1,-1}[m_1, m_d^{i2}, m_q^p] + \\
& \frac{1}{36} m_1 g_1^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{3,1,1,-1}[m_1, m_d^{i2}, m_q^p] - \frac{1}{72} m_1 g_1^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{2,2,1,-1}[m_1, m_q^p, m_d^{i2}] + \\
& \frac{1}{18} m_1 g_1^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{2,1,1,0}[m_1, m_q^p, m_u^{i1}] + \frac{1}{36} m_1 g_1^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{2,2,1,-1}[m_1, m_q^p, m_u^{i1}] - \\
& \frac{1}{18} m_1 g_1^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{3,1,1,-1}[m_1, m_q^p, m_u^{i1}] - \frac{1}{36} m_1 g_1^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{2,2,1,-1}[m_1, m_u^{i1}, m_q^p] - \\
& \frac{1}{12} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_d^{i2}] + \frac{1}{8} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_q^p] + \\
& \frac{1}{8} g_1^2 \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_q^p] - \frac{1}{6} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_u^{i1}] + \\
& \frac{9}{4} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,-1}[m_2, m_q^p, \tilde{\mu}] - \frac{3}{2} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,1,-2}[m_2, m_q^p, \tilde{\mu}] - \\
& \frac{9}{8} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_q^p] - \frac{3}{8} g_2^2 \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^p] + \\
& \frac{2}{3} m_3 g_3^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{2,1,1,0}[m_3, m_d^{i2}, m_q^p] - \frac{1}{3} m_3 g_3^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{2,2,1,-1}[m_3, m_d^{i2}, m_q^p] - \\
& \frac{2}{3} m_3 g_3^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{3,1,1,-1}[m_3, m_d^{i2}, m_q^p] + \frac{1}{3} m_3 g_3^2 \overline{y_u}^{pi1} a_d^{pi2} LF_{2,2,1,-1}[m_3, m_q^p, m_d^{i2}] + \\
& \frac{2}{3} m_3 g_3^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{2,1,1,0}[m_3, m_q^p, m_u^{i1}] + \frac{1}{3} m_3 g_3^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{2,2,1,-1}[m_3, m_q^p, m_u^{i1}] - \\
& \frac{2}{3} m_3 g_3^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{3,1,1,-1}[m_3, m_q^p, m_u^{i1}] - \frac{1}{3} m_3 g_3^2 y_d^{pi2} \overline{a_u}^{pi1} LF_{2,2,1,-1}[m_3, m_u^{i1}, m_q^p] + \\
& \frac{1}{8} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{2,2,1,-1}[m_d^r, \tilde{\mu}, m_q^s] - \frac{1}{8} \tilde{\mu}^2 y_d^{ri2} \overline{y_u}^{pi1} \overline{y_u}^{rs} y_u^{ps} \\
& LF_{2,2,1,-1}[m_q^r, \tilde{\mu}, m_u^s] - \frac{1}{8} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{2,2,1,-1}[m_q^s, \tilde{\mu}, m_d^r] + \\
& \frac{1}{8} \tilde{\mu}^2 y_d^{ri2} \overline{y_u}^{pi1} \overline{y_u}^{rs} y_u^{ps} LF_{2,2,1,-1}[m_u^s, \tilde{\mu}, m_q^r] + \frac{1}{4} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_d^{i2}] - \\
& \frac{1}{6} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_d^{i2}] + \frac{1}{8} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_q^p] - \\
& \frac{1}{4} g_1^2 \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_q^p] + \frac{1}{2} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_u^{i1}] - \\
& \frac{1}{3} g_1^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_u^{i1}] - \frac{3}{8} g_2^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,-1}[\tilde{\mu}, m_2, m_q^p] + \\
& \frac{3}{4} g_2^2 \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pi1} LF_{2,1,1,0}[\tilde{\mu}, m_2, m_q^p] + \frac{1}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{2,1,1,0}[\tilde{\mu}, m_d^r, m_q^s] - \\
& \frac{1}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pi2} y_d^{sr} \overline{y_u}^{si1} LF_{3,1,1,-1}[\tilde{\mu}, m_d^r, m_q^s] + \frac{1}{4} \tilde{\mu}^2 y_d^{ri2} \overline{y_u}^{pi1} \overline{y_u}^{rs} y_u^{ps} \\
& LF_{2,1,1,0}[\tilde{\mu}, m_q^r, m_u^s] - \frac{1}{4} \tilde{\mu}^2 y_d^{ri2} \overline{y_u}^{pi1} \overline{y_u}^{rs} y_u^{ps} LF_{3,1,1,-1}[\tilde{\mu}, m_q^r, m_u^s] - \\
& \frac{1}{12} g_1^2 \overline{y_u}^{pi1} (m_1 a_d^{pi2} + \tilde{\mu}^2 y_d^{pi2}) LF_{1,1,1,1,0}[m_1, m_d^{i2}, m_q^p, \tilde{\mu}] + \\
& \frac{1}{24} g_1^2 \overline{y_u}^{pi1} (-m_1 a_d^{pi2} + \tilde{\mu}^2 y_d^{pi2}) LF_{2,1,1,1,-1}[m_1, m_d^{i2}, m_q^p, \tilde{\mu}] - \\
& \frac{1}{6} g_1^2 y_d^{pi2} (m_1 \overline{a_u}^{pi1} + \tilde{\mu}^2 \overline{y_u}^{pi1}) LF_{1,1,1,1,0}[m_1, m_q^p, m_u^{i1}, \tilde{\mu}] + \\
& \frac{1}{12} g_1^2 y_d^{pi2} (-m_1 \overline{a_u}^{pi1} + \tilde{\mu}^2 \overline{y_u}^{pi1}) LF_{2,1,1,1,-1}[m_1, m_q^p, m_u^{i1}, \tilde{\mu}] + \\
& \frac{1}{24} g_1^2 \overline{y_u}^{pi1} (-m_1 a_d^{pi2} + \tilde{\mu}^2 y_d^{pi2}) LF_{2,1,1,1,-1}[m_d^{i2}, m_1, m_q^p, \tilde{\mu}] - \\
& \frac{1}{9} g_1^2 (a_d^{pi2} \overline{a_u}^{pi1} + \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pi1}) LF_{2,1,1,1,-1}[m_q^p, m_1, m_d^{i2}, m_u^{i1}] + \\
& \frac{1}{24} g_1^2 \overline{y_u}^{pi1} (m_1 a_d^{pi2} - \tilde{\mu}^2 y_d^{pi2}) LF_{2,1,1,1,-1}[m_q^p, m_1, m_d^{i2}, \tilde{\mu}] + \\
& \frac{1}{12} g_1^2 y_d^{pi2} (m_1 \overline{a_u}^{pi1} - \tilde{\mu}^2 \overline{y_u}^{pi1}) LF_{2,1,1,1,-1}[m_q^p, m_1, m_u^{i1}, \tilde{\mu}] + \\
& \frac{2}{3} g_3^2 (a_d^{pi2} \overline{a_u}^{pi1} + \tilde{\mu}^2 y_d^{pi2} \overline{y_u}^{pi1}) LF_{2,1,1,1,-1}[m_q^p, m_3, m_d^{i2}, m_u^{i1}] + \\
& \frac{1}{12} g_1^2 y_d^{pi2} (-m_1 \overline{a_u}^{pi1} + \tilde{\mu}^2 \overline{y_u}^{pi1}) LF_{2,1,1,1,-1}[m_u^{i1}, m_1, m_q^p, \tilde{\mu}] + \\
& \frac{1}{24} g_1^2 \overline{y_u}^{pi1} (m_1 a_d^{pi2} - \tilde{\mu}^2 y_d^{pi2}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_d^{i2}, m_q^p] + \\
& \frac{1}{12} g_1^2 y_d^{pi2} (m_1 \overline{a_u}^{pi1} - \tilde{\mu}^2 \overline{y_u}^{pi1}) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_q^p, m_u^{i1}])
\end{aligned}$$